

c. Solve the equation $z^4 + 16p^2i = 0$, where p is real.

Write your solutions in polar form, in terms of p .

$$z^4 = -16p^2i$$

$$z^4 = 16p^2 \operatorname{cis} \left(\frac{3\pi}{2} + 2k\pi \right)$$

$$z = 2\sqrt{p} \operatorname{cis} \left(\frac{3\pi}{8} + \frac{k\pi}{2} \right)$$

$$z_1 = 2\sqrt{p} \operatorname{cis} \left(\frac{3\pi}{8} \right)$$

$$z_2 = 2\sqrt{p} \operatorname{cis} \left(\frac{7\pi}{8} \right)$$

$$z_3 = 2\sqrt{p} \operatorname{cis} \left(\frac{-5\pi}{8} \right)$$

$$z_4 = 2\sqrt{p} \operatorname{cis} \left(\frac{-\pi}{8} \right)$$